Local Nonlinear Projection-Based Reduced-Order Models

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Setup for Paper Campaign

Problem: 2D Burgers, mesh 250×250.

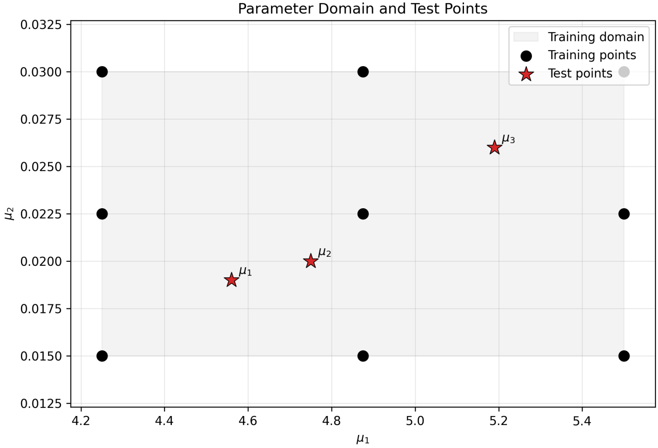
Test points:

$$\boldsymbol{\mu}_1 = (4.56, 0.019), \quad \boldsymbol{\mu}_2 = (4.75, 0.020), \quad \boldsymbol{\mu}_3 = (5.19, 0.026).$$

Methods included:

- Global HPROM (linear POD)
- Global HQPROM (quadratic manifold)
- Global HPROM-GPR
- Local HPROM
- Local HQPROM
- Local HPROM-GPR

Parameter Domain and Test Points



Global ROM Storyline: LSPG + Gauss-Newton

Given the implicit HDM residual $\mathbf{r}(\mathbf{u}) = \mathbf{0}$, define a manifold

$$\mathbf{u} \approx \mathbf{u}_{\mathrm{ref}} + \widetilde{\mathbf{u}}(\mathbf{q}).$$

LSPG solves at each time step:

$$\mathbf{q}^{\star} = \arg\min_{\mathbf{q}} \ \frac{1}{2} \left\| \mathbf{r}(\mathbf{u}_{\mathrm{ref}} + \widetilde{\mathbf{u}}(\mathbf{q})) \right\|_2^2 .$$

With Gauss-Newton (current iterate $\mathbf{q}^{(k)}$), the increment $\Delta\mathbf{q}$ is obtained from

$$\left(\boldsymbol{\Psi}^T \mathbf{J}^T \mathbf{J} \boldsymbol{\Psi} \right) \Delta\mathbf{q} = -\boldsymbol{\Psi}^T \mathbf{J}^T \mathbf{r},$$

where

$$\boldsymbol{\Psi} = \frac{\partial \widetilde{\mathbf{u}}}{\partial \mathbf{q}}, \qquad \mathbf{J} = \frac{\partial \mathbf{r}}{\partial \mathbf{u}}, \qquad \mathbf{q}^{(k+1)} = \mathbf{q}^{(k)} + \Delta\mathbf{q}.$$

Global Linear HPROM

State ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\mathrm{ref}} + \mathbf{V}\mathbf{q}, \qquad \mathbf{V} \in \mathbb{R}^{N \times n},\ n = 96.$$

Hence

$$\widetilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}\mathbf{q}, \qquad \boldsymbol{\Psi} = \frac{\partial \widetilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V}.$$

Gauss-Newton system:

$$\left(\mathbf{V}^T \mathbf{J}^T \mathbf{J} \mathbf{V}\right) \Delta\mathbf{q} = -\mathbf{V}^T \mathbf{J}^T \mathbf{r}.$$

Quadratic-manifold ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\mathrm{ref}} + \mathbf{V}\mathbf{q} + \mathbf{H}\left(\mathbf{q} \otimes \mathbf{q}\right), \qquad n = 39.$$

Thus

$$\widetilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}\mathbf{q} + \mathbf{H}\left(\mathbf{q} \otimes \mathbf{q}\right),$$

and the tangent depends on the current reduced state:

$$\mathbf{\Psi}(\mathbf{q}) = \frac{\partial \widetilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V} + \frac{\partial}{\partial \mathbf{q}}[\mathbf{H}\left(\mathbf{q} \otimes \mathbf{q}\right)].$$

Split basis and reduced coordinates:

$$\mathbf{V}_{\mathrm{tot}} = [\mathbf{V}, \bar{\mathbf{V}}], \qquad \mathbf{q}_{\mathrm{tot}} = \begin{bmatrix} \mathbf{q} \\ \bar{\mathbf{q}} \end{bmatrix}.$$

$$\bar{\mathbf{q}} = \mathcal{N}_{\mathrm{GPR}}(\mathbf{q}), \qquad n = 20, \; \bar{n} = 131.$$

State ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\mathrm{ref}} + \widetilde{\mathbf{u}}(\mathbf{q}),$$

$$\widetilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}_{\mathrm{tot}}\mathbf{q}_{\mathrm{tot}} = \mathbf{V}\mathbf{q} + \bar{\mathbf{V}}\,\mathcal{N}_{\mathrm{GPR}}(\mathbf{q}).$$

Effective tangent for Gauss-Newton:

$$\boldsymbol{\Psi}(\mathbf{q}) = \frac{\partial \widetilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V} + \bar{\mathbf{V}}\frac{\partial \mathcal{N}_{\mathrm{GPR}}}{\partial \mathbf{q}}.$$

Partition the training snapshots into clusters $k = 1, \ldots, N_c$ and build one local trial manifold per cluster:

$$\mathbf{u} \approx \Phi_k(\mathbf{q}_k).$$

Examples used in this work:

$$\Phi_k^{\mathrm{lin}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k\mathbf{q},$$

$$\Phi_k^{\mathrm{quad}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k\mathbf{q} + \mathbf{H}_k(\mathbf{q} \otimes \mathbf{q}),$$

$$\Phi_k^{\mathrm{gpr}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k\mathbf{q} + \bar{\mathbf{V}}_k\,\mathcal{N}_{\mathrm{GPR},k}(\mathbf{q}).$$

Online workflow

1. Select the active cluster.
2. If the cluster changes, transfer the current state to the new local manifold.
3. Solve the local LSPG problem in the active manifold.

Cluster Selection: Final Form (Linear)

Shared nearest-centroid selector (used in all local models):

$$k^{+} = \arg \min_{\ell=1,\ldots,N_c} \|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2, \qquad \mathbf{u}^s = \Phi_{k^-}(\mathbf{q}_{k^-}).$$

For the linear local manifold $\Phi_{k^-}^{\mathrm{lin}}(\mathbf{q}) = \mathbf{u}_{0,k^-} + \mathbf{V}_{k^-}\mathbf{q}$:

$$k^{+} = \arg \min_{\ell=1,\ldots,N_c} S_{k^-,\ell}^{\mathrm{lin}}(\mathbf{q}_{k^-}),$$

$$S_{k^-,\ell}^{\mathrm{lin}}(\mathbf{q}) = 2\,\mathbf{g}_{k^-,\ell}^{T}\mathbf{q} + d_{k^-,\ell},$$

with offline-precomputed terms

$$\mathbf{g}_{k^-,\ell} := \mathbf{V}_{k^-}^{T}(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}), \qquad d_{k^-,\ell} := \|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2.$$

HQPROM selector

$$k^{+} = \arg\min_{\ell} S^{\text{quad}}_{k-,\ell}(\mathbf{q}_{k-}),$$

$$S^{\text{quad}}_{k-,\ell}(\mathbf{q}) = 2\,\mathbf{g}^{T}_{k-,\ell}\mathbf{q} + 2\,\mathbf{m}^{T}_{k-,\ell}(\mathbf{q}\otimes\mathbf{q}) + d_{k-,\ell},$$

$$\mathbf{m}^{T}_{k-,\ell}(\mathbf{q}\otimes\mathbf{q}) := (\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell})^{T}\mathbf{H}_{k-}(\mathbf{q}\otimes\mathbf{q}).$$

HPROM-GPR selector

$$k^{+} = \arg\min_{\ell} S^{\text{gpr}}_{k-,\ell}(\mathbf{q}_{k-}),$$

$$S^{\text{gpr}}_{k-,\ell}(\mathbf{q}) = 2\,\mathbf{g}^{T}_{k-,\ell}\mathbf{q} + 2\,\bar{\mathbf{g}}^{T}_{k-,\ell}\mathcal{N}_{\text{GPR},k-}(\mathbf{q}) + d_{k-,\ell}.$$

$$\bar{\mathbf{q}}_{k-} := \mathcal{N}_{\text{GPR},k-}(\mathbf{q}_{k-}), \qquad \bar{\mathbf{g}}_{k-,\ell} := \bar{\mathbf{V}}^{T}_{k-}(\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}).$$

Cluster Change: Linear and Nonlinear Transfers

$$k^+ = k^- \Rightarrow \text{no transfer}, \qquad k^+ \neq k^- \Rightarrow \text{solve transfer problem.}$$

Transferred state:

$$\mathbf{u}^s = \Phi_{k^-}(\mathbf{q}_{k^-}).$$

Linear local target

$$\mathbf{q}_{k^+}^{\mathrm{lin}} = \arg\min_{\mathbf{q}} \left\| \mathbf{u}^s - \left(\mathbf{u}_{0,k^+} + \mathbf{V}_{k^+}\mathbf{q} \right) \right\|_2^2,$$

$$\text{if } \mathbf{V}_{k^+}^T \mathbf{V}_{k^+} = \mathbf{I}, \quad \mathbf{q}_{k^+}^{\mathrm{lin}} = \mathbf{V}_{k^+}^T (\mathbf{u}^s - \mathbf{u}_{0,k^+}).$$

Nonlinear local targets (HQPROM / HPROM-GPR)

$$\mathbf{q}_{k^+}^{\mathrm{nl}} = \arg\min_{\mathbf{q}} \left\| \mathbf{u}^s - \Phi_{k^+}^{\mathrm{nl}}(\mathbf{q}) \right\|_2^2.$$

For nonlinear local manifolds, define

$$\mathbf{r}_{\mathrm{tr}}(\mathbf{q}) := \Phi_{k^+}(\mathbf{q}) - \mathbf{u}^s, \qquad \mathbf{J}_\Phi(\mathbf{q}) := \frac{\partial \Phi_{k^+}}{\partial \mathbf{q}}.$$

Gauss–Newton at iterate $\mathbf{q}^{(j)}$:

$$\left(\mathbf{J}_\Phi^T \mathbf{J}_\Phi\right) \Delta\mathbf{q} = -\mathbf{J}_\Phi^T \mathbf{r}_{\mathrm{tr}}(\mathbf{q}^{(j)}), \qquad \mathbf{q}^{(j+1)} = \mathbf{q}^{(j)} + \Delta\mathbf{q}.$$

Online Performance (250 $\times$ 250): Avg/Max Error over 3 Test Points

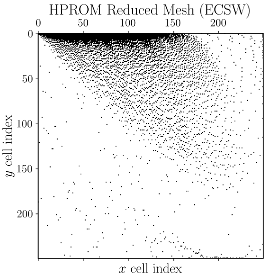
Computational model	n (N for HDM)	$\bar{n}$	N_e	avg $\mathbb{RE}_{2,\mathbf{u}}$ (%)	max $\mathbb{RE}_{2,\mathbf{u}}$ (%)	Wall-clock time (min)	Speedup
HDM	125 000	–	62 500	–	–	12.512	–
HPROM	96	–	4 824	1.030	1.096	0.743	16.841
HQPROM	39	–	3 505	0.871	0.933	0.888	14.084
HPROM-GPR	20	131	1 801	0.691	0.845	0.371	33.698
Local HPROM ($N_c = 3$)	35–48	–	3 649	0.827	0.854	0.377	33.168
Local HQPROM ($N_c = 3$)	11–15	–	1 164	0.716	0.844	0.311	40.259
Local HPROM-GPR ($N_c = 3$)	9	60–97	811	0.713	0.789	0.282	44.379

Speedup is computed with respect to the HDM runtime, $t_{\mathrm{HDM}} = 750.702$ s.

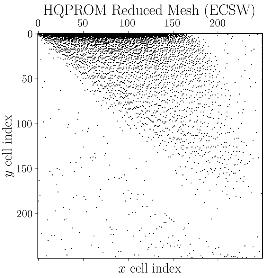
N_e denotes the number of sampled ECSW elements ($N_e = 62{,}500$ for HDM/full mesh).

Local ROM results shown here use $N_c = 3$ clusters; the $N_c = 10$ results are reproduced in the appendix.

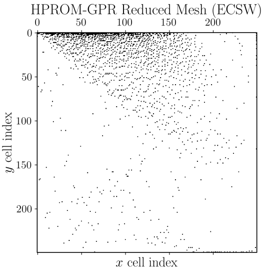
Global Reduced Meshes (ECSW)



HPROM (N_e = 4,824)

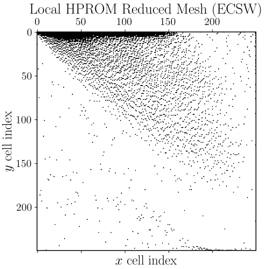


HQPROM (N_e = 3,505)

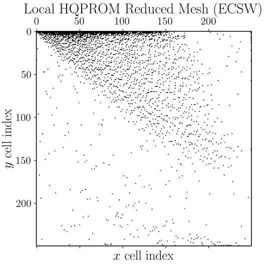


HPROM-GPR (N_e = 1,801)

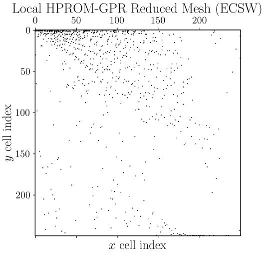
Local Reduced Meshes (ECSW), $N_c = 3$



Local HPROM (N_e = 3,649)

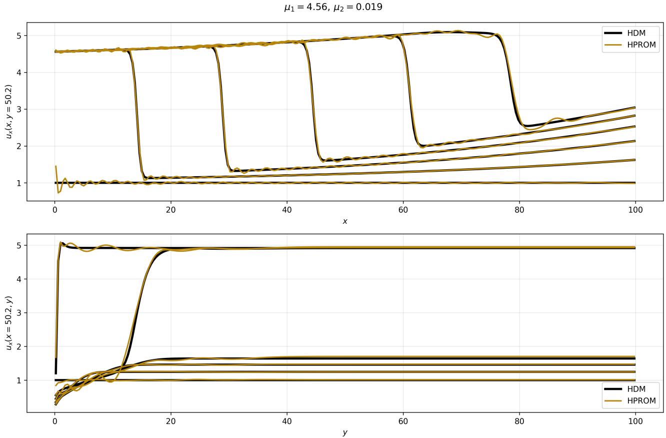


Local HQPROM (N_e = 1,164)

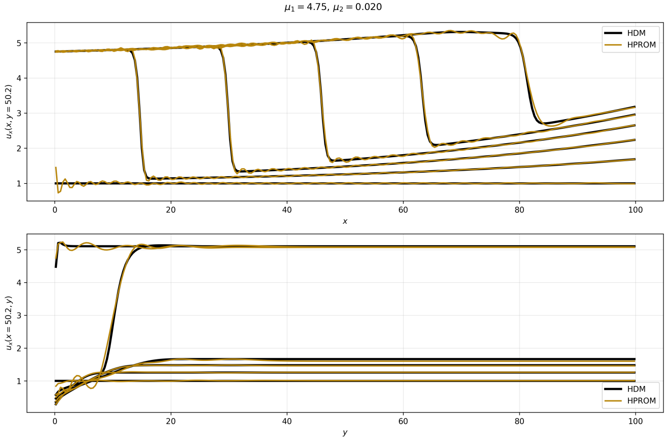


Local HPROM-GPR (N_e = 811)

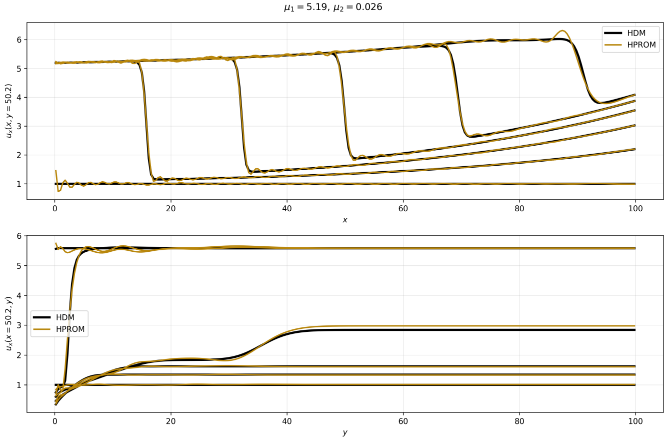
HPROM vs HDM: $\boldsymbol{\mu} = (4.56,\ 0.019)$



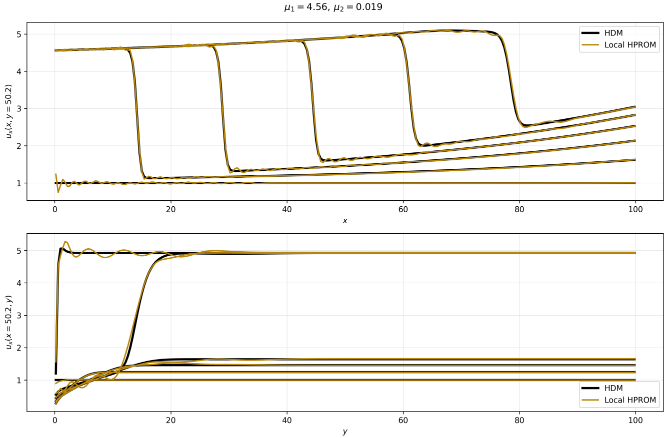
HPROM vs HDM: $\boldsymbol{\mu} = (4.75, 0.020)$



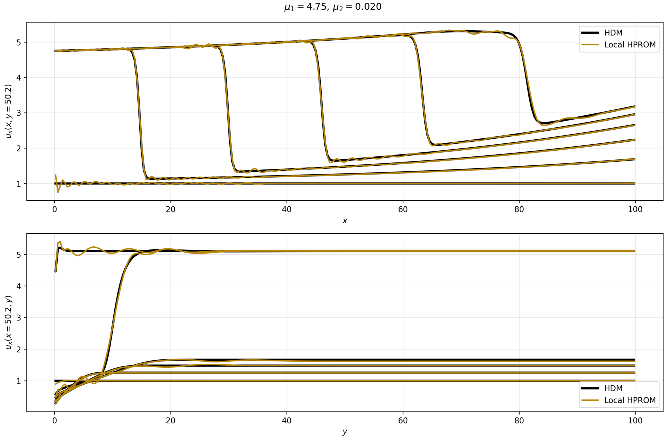
HPROM vs HDM: $\boldsymbol{\mu} = (5.19,\ 0.026)$



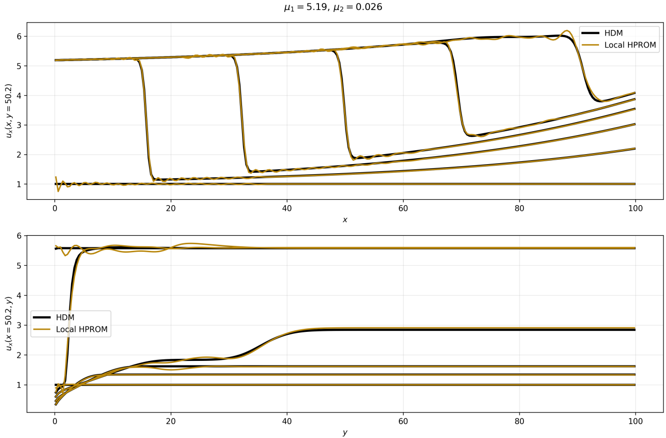
Local HPROM ($N_c = 3$) vs HDM: $\boldsymbol{\mu} = (4.56, 0.019)$



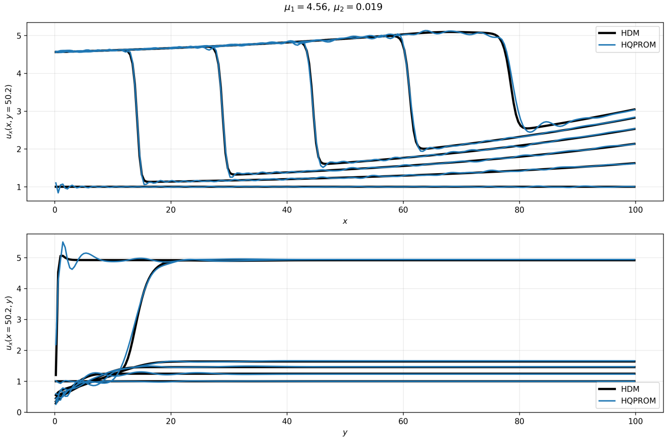
Local HPROM ($N_c = 3$) vs HDM: $\boldsymbol{\mu} = (4.75, 0.020)$



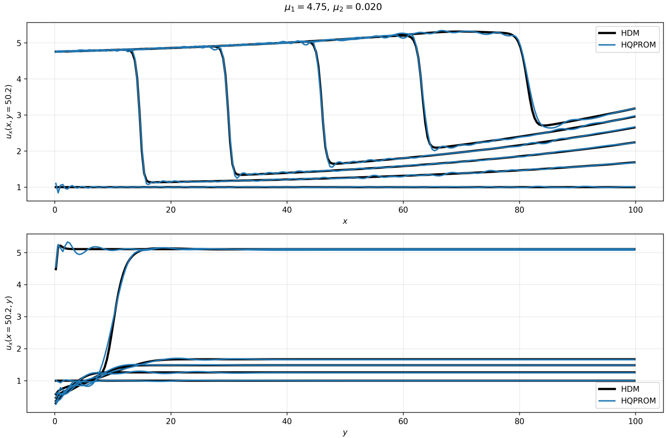
Local HPROM ($N_c = 3$) vs HDM: $\boldsymbol{\mu} = (5.19, 0.026)$



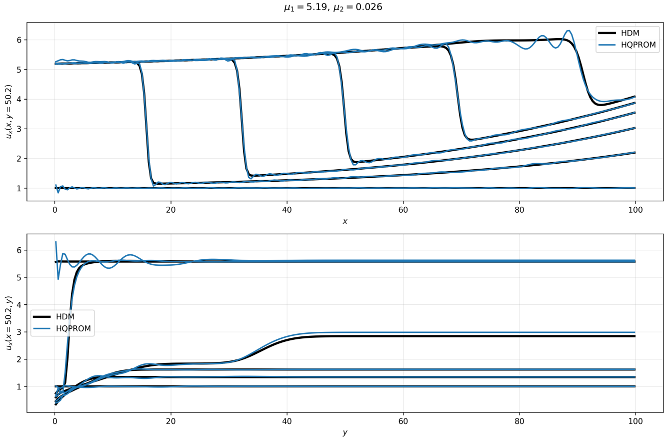
HQPROM vs HDM: $\boldsymbol{\mu} = (4.56,\ 0.019)$



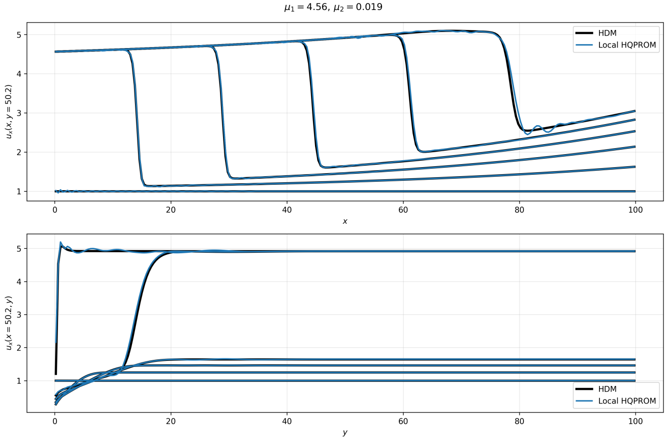
HQPROM vs HDM: $\boldsymbol{\mu} = (4.75,\ 0.020)$



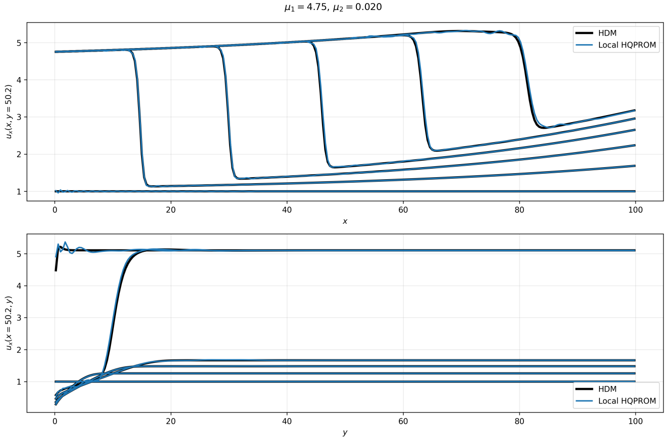
HQPROM vs HDM: $\boldsymbol{\mu} = (5.19,\ 0.026)$



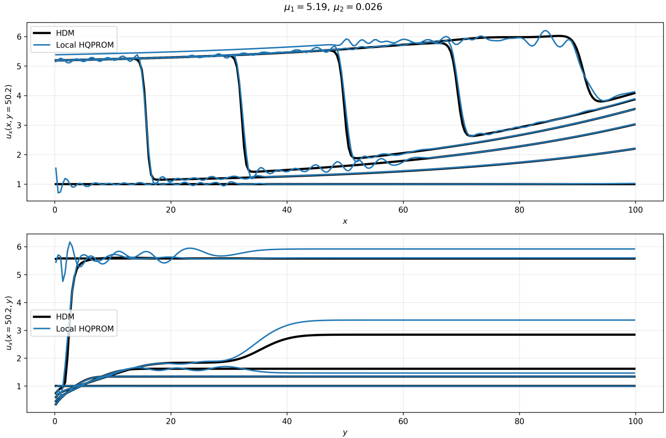
Local HQPROM ($N_c = 3$) vs HDM: $\boldsymbol{\mu} = (4.56, 0.019)$



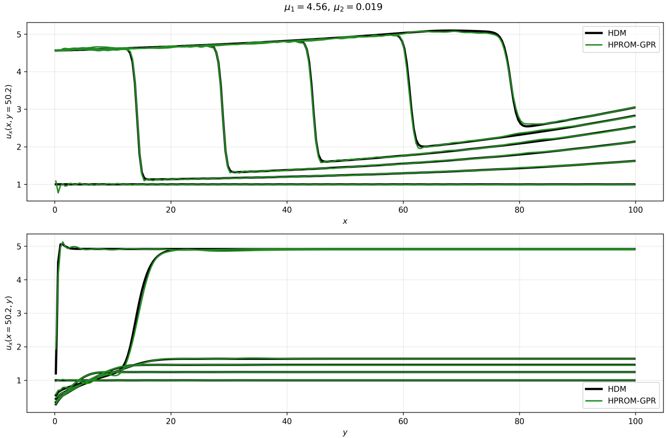
Local HQPROM ($N_c = 3$) vs HDM: $\boldsymbol{\mu} = (4.75, 0.020)$



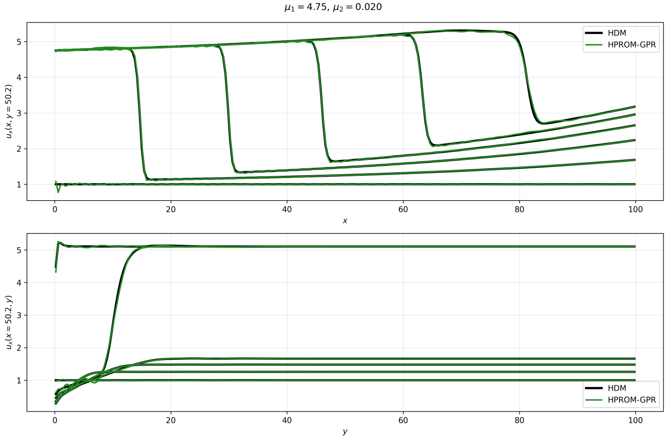
Local HQPROM ($N_c = 3$) vs HDM: $\boldsymbol{\mu} = (5.19, 0.026)$



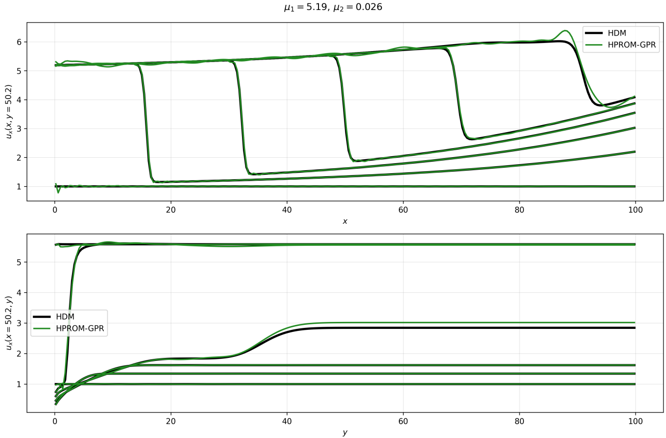
HPROM-GPR vs HDM: $\boldsymbol{\mu} = (4.56,\ 0.019)$



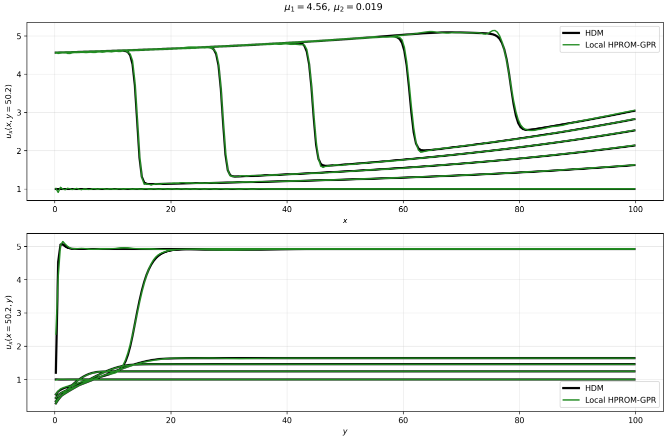
HPROM-GPR vs HDM: $\boldsymbol{\mu} = (4.75,\ 0.020)$



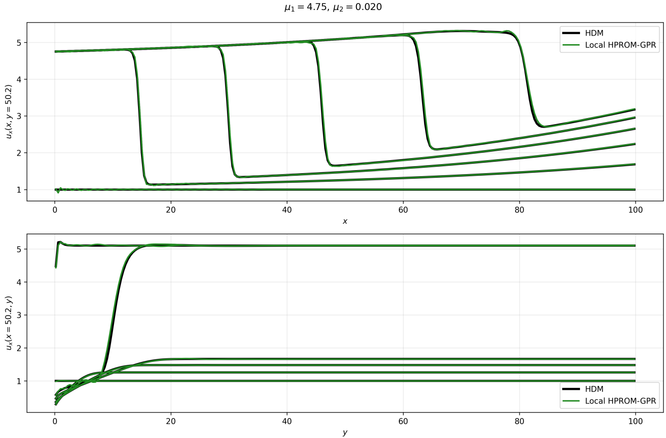
HPROM-GPR vs HDM: $\boldsymbol{\mu} = (5.19,\ 0.026)$



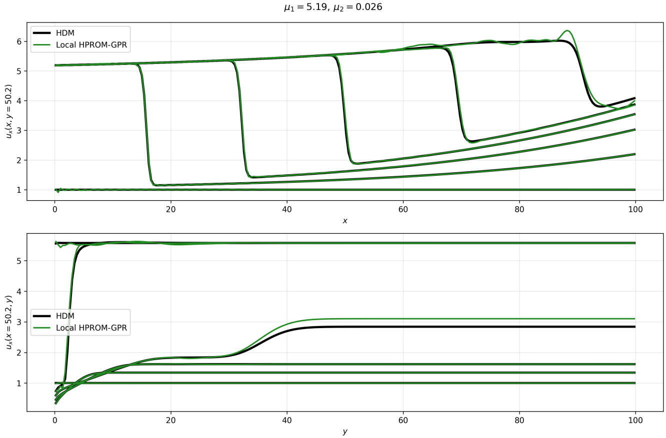
Local HPROM-GPR ($N_c = 3$) vs HDM: $\boldsymbol{\mu} = (4.56, 0.019)$



Local HPROM-GPR ($N_c = 3$) vs HDM: $\boldsymbol{\mu} = (4.75, 0.020)$



Local HPROM-GPR ($N_c = 3$) vs HDM: $\boldsymbol{\mu} = (5.19, 0.026)$



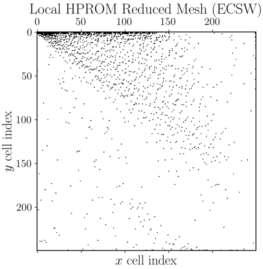
Appendix: Local ROM Results at $N_c = 10$

Computational model	n	$\bar{n}$	N_e	avg $\mathbb{RE}_{2,\mathbf{u}}$ (%)	max $\mathbb{RE}_{2,\mathbf{u}}$ (%)	Wall-clock time (min)	Speedup
Local HPROM ($N_c = 10$)	14–23	–	1 522	0.867	0.956	0.198	63.257
Local HQPROM ($N_c = 10$)	8–12	–	871	0.572	0.929	0.263	47.497
Local HPROM-GPR ($N_c = 10$)	6	24–47	541	0.604	0.801	0.244	51.261

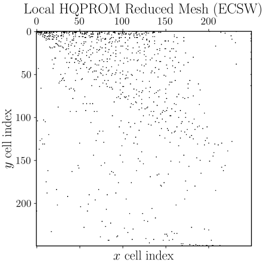
Speedup is computed with respect to the HDM runtime, $t_{\mathrm{HDM}} = 750.702$ s.

Global-model rows are unaffected by N_c and are reproduced in the main results table.

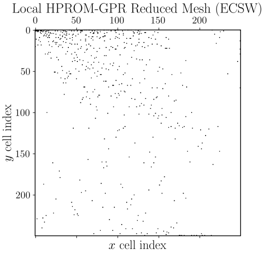
Appendix: Local Reduced Meshes (ECSW), $N_c = 10$



Local HPROM (N_e = 1,522)

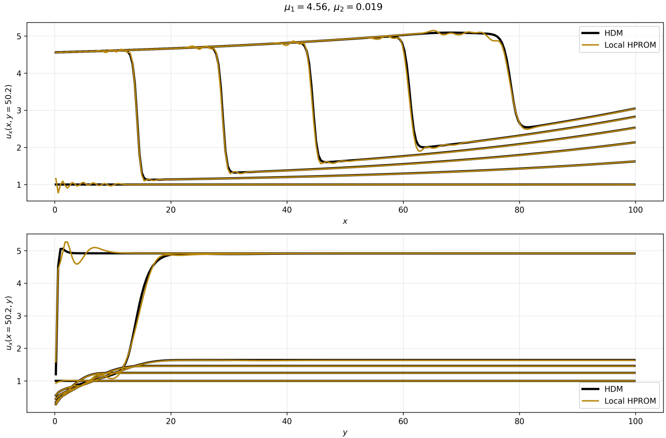


Local HQPROM (N_e = 871)

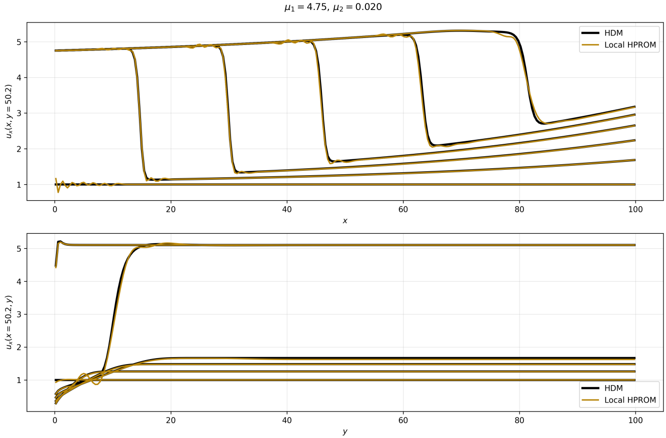


Local HPROM-GPR (N_e = 541)

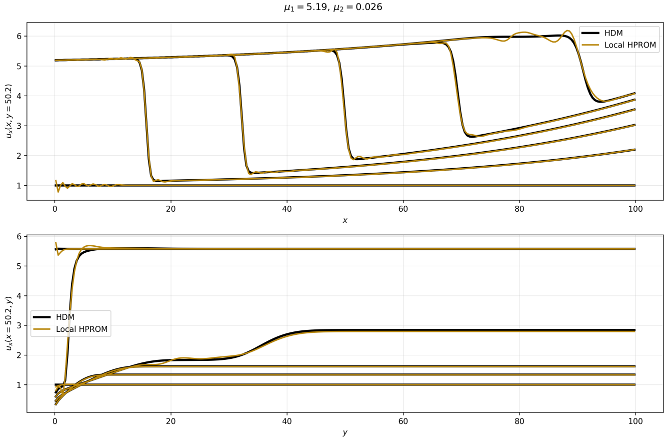
Appendix: Local HPROM ($N_c = 10$) vs HDM: $\boldsymbol{\mu} = (4.56,\ 0.019)$



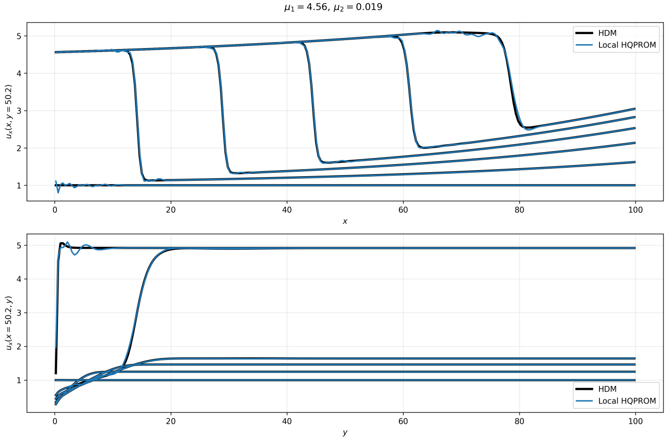
Appendix: Local HPROM ($N_c = 10$) vs HDM: $\boldsymbol{\mu} = (4.75, 0.020)$



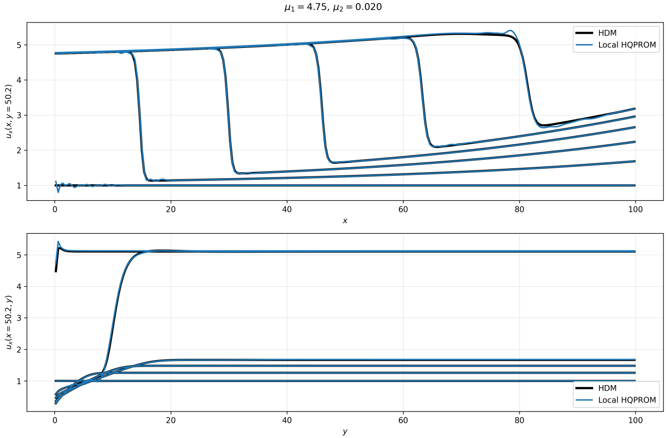
Appendix: Local HPROM ($N_c = 10$) vs HDM: $\boldsymbol{\mu} = (5.19,\ 0.026)$



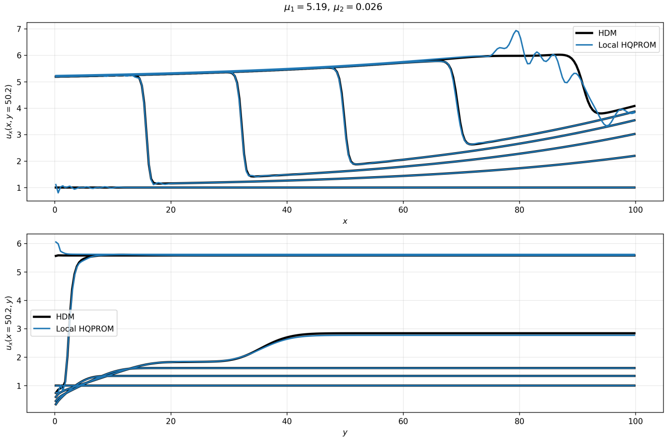
Appendix: Local HQPROM ($N_c = 10$) vs HDM: $\boldsymbol{\mu} = (4.56, 0.019)$



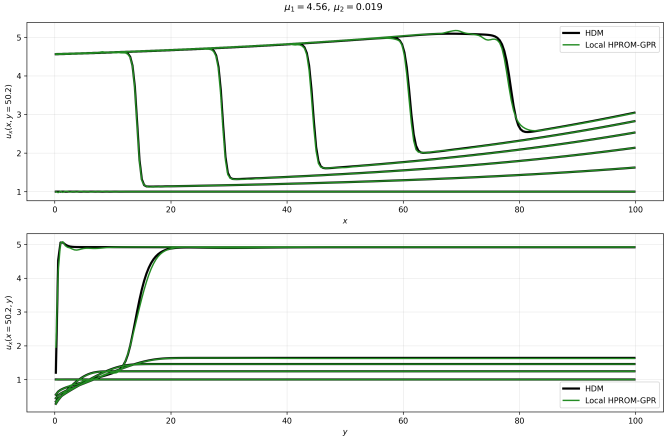
Appendix: Local HQPROM ($N_c = 10$) vs HDM: $\boldsymbol{\mu} = (4.75, 0.020)$



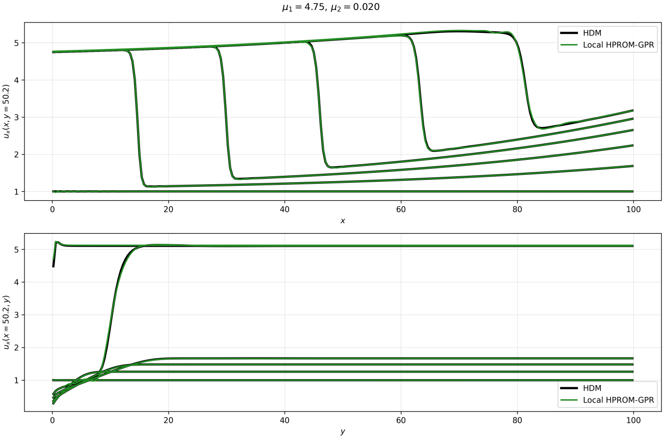
Appendix: Local HQPROM ($N_c = 10$) vs HDM: $\boldsymbol{\mu} = (5.19, 0.026)$



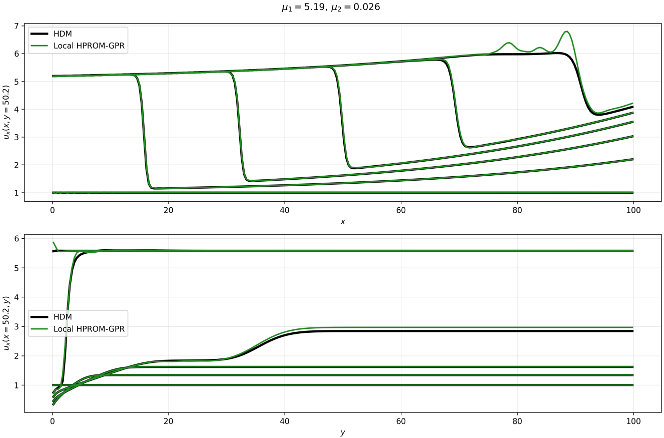
Appendix: Local HPROM-GPR ($N_c = 10$) vs HDM: $\boldsymbol{\mu} = (4.56, 0.019)$



Appendix: Local HPROM-GPR ($N_c = 10$) vs HDM: $\boldsymbol{\mu} = (4.75, 0.020)$



Appendix: Local HPROM-GPR ($N_c = 10$) vs HDM: $\boldsymbol{\mu} = (5.19, 0.026)$



Appendix: Linear Selector Expansion

$$\mathbf{u}^s = \mathbf{u}_{0,k-} + \mathbf{V}_{k-}\mathbf{q}_{k-}, \qquad k^+ = \arg\min_{\ell} \|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2.$$

$$\text{Let } \mathbf{a}_\ell := \mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}, \quad \mathbf{b} := \mathbf{V}_{k-}\mathbf{q}_{k-}.$$

$$\|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2 = \|\mathbf{a}_\ell + \mathbf{b}\|_2^2 = \|\mathbf{a}_\ell\|_2^2 + \|\mathbf{b}\|_2^2 + 2\,\mathbf{a}_\ell^T\mathbf{b}.$$

$$\|\mathbf{b}\|_2^2 = \|\mathbf{V}_{k-}\mathbf{q}_{k-}\|_2^2 \text{ is independent of } \ell.$$

$$\Rightarrow\ k^+ = \arg\min_{\ell}\left[\|\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}\|_2^2 + 2(\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell})^T\mathbf{V}_{k-}\mathbf{q}_{k-}\right].$$

Precompute

$$\mathbf{g}_{k-,\ell} := \mathbf{V}_{k-}^T(\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}), \qquad d_{k-,\ell} := \|\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}\|_2^2,$$

which yields

$$k^+ = \arg\min_{\ell}\left(2\,\mathbf{g}_{k-,\ell}^T\mathbf{q}_{k-} + d_{k-,\ell}\right).$$

$$\mathbf{u}^s = \mathbf{u}_{0,k-} + \mathbf{V}_{k-}\mathbf{q}_{k-} + \mathbf{H}_{k-}(\mathbf{q}_{k-} \otimes \mathbf{q}_{k-}).$$

$$\text{Let } \mathbf{a}_\ell := \mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}, \quad \mathbf{b} := \mathbf{V}_{k-}\mathbf{q}_{k-}, \quad \mathbf{c} := \mathbf{H}_{k-}(\mathbf{q}_{k-} \otimes \mathbf{q}_{k-}).$$

$$k^+ = \arg\min_\ell \|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2 = \arg\min_\ell \|\mathbf{a}_\ell + \mathbf{b} + \mathbf{c}\|_2^2.$$

$$\|\mathbf{a}_\ell + \mathbf{b} + \mathbf{c}\|_2^2 = \|\mathbf{a}_\ell\|_2^2 + \|\mathbf{b}\|_2^2 + \|\mathbf{c}\|_2^2 + 2\,\mathbf{a}_\ell^T\mathbf{b} + 2\,\mathbf{a}_\ell^T\mathbf{c} + 2\,\mathbf{b}^T\mathbf{c}.$$

$$\|\mathbf{b}\|_2^2,\ \|\mathbf{c}\|_2^2,\ 2\,\mathbf{b}^T\mathbf{c} \text{ are independent of } \ell.$$

$$\Rightarrow\ k^+ = \arg\min_\ell \Big[\|\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}\|_2^2$$

$$+2(\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell})^T\mathbf{V}_{k-}\mathbf{q}_{k-} + 2(\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell})^T\mathbf{H}_{k-}(\mathbf{q}_{k-} \otimes \mathbf{q}_{k-})\Big].$$

Define

$$\mathbf{m}_{k-,\ell}^T(\mathbf{q} \otimes \mathbf{q}) := (\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell})^T\mathbf{H}_{k-}(\mathbf{q} \otimes \mathbf{q}),$$

then

$$k^+ = \arg\min_\ell \left(2\,\mathbf{g}_{k-,\ell}^T\mathbf{q}_{k-} + 2\,\mathbf{m}_{k-,\ell}^T(\mathbf{q}_{k-} \otimes \mathbf{q}_{k-}) + d_{k-,\ell}\right).$$

Appendix: HPROM-GPR Selector Expansion

$$\mathbf{u}^s = \mathbf{u}_{0,k^-} + \mathbf{V}_{k^-}\mathbf{q}_{k^-} + \bar{\mathbf{V}}_{k^-}\bar{\mathbf{q}}_{k^-}, \qquad \bar{\mathbf{q}}_{k^-} = \mathcal{N}_{\mathrm{GPR},k^-}(\mathbf{q}_{k^-}).$$

$$\text{Let } \mathbf{a}_\ell := \mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}, \quad \mathbf{b} := \mathbf{V}_{k^-}\mathbf{q}_{k^-}, \quad \mathbf{c} := \bar{\mathbf{V}}_{k^-}\bar{\mathbf{q}}_{k^-}.$$

$$\|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2 = \|\mathbf{a}_\ell + \mathbf{b} + \mathbf{c}\|_2^2$$

$$= \|\mathbf{a}_\ell\|_2^2 + \|\mathbf{b}\|_2^2 + \|\mathbf{c}\|_2^2 + 2\,\mathbf{a}_\ell^T\mathbf{b} + 2\,\mathbf{a}_\ell^T\mathbf{c} + 2\,\mathbf{b}^T\mathbf{c}.$$

$$\|\mathbf{b}\|_2^2,\ \|\mathbf{c}\|_2^2,\ 2\,\mathbf{b}^T\mathbf{c} \text{ are independent of } \ell.$$

$$\Rightarrow\ k^+ = \arg\min_\ell \left[\|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2 + 2(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell})^T\mathbf{V}_{k^-}\mathbf{q}_{k^-} + 2(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell})^T\bar{\mathbf{V}}_{k^-}\bar{\mathbf{q}}_{k^-}\right].$$

Hence, with

$$\bar{\mathbf{g}}_{k^-,\ell} := \bar{\mathbf{V}}_{k^-}^T(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}),$$

$$k^+ = \arg\min_\ell \left(2\,\mathbf{g}_{k^-,\ell}^T\mathbf{q}_{k^-} + 2\,\bar{\mathbf{g}}_{k^-,\ell}^T\bar{\mathbf{q}}_{k^-} + d_{k^-,\ell}\right).$$